CT30A8922 - User Experience Design

Responsible Teacher: Assoc. Prof. Annika Wolff

Assignment 2

UX Design Portfolio: Enhancing Freebike Customer Support

Word count: 3567

Group - Lahti4ever

Authors of the Report:

Artturi Sivén

Trieu Huynh Ba Nguyen

Rihards Gailis

Nazmul Khan

Declaration of AI Usage

AI Tools Utilized:

We/I in line with LUT AI usage recommendations and policy (https://elut.lut.fi/en/completing-studies/rules-and-regulations/ai-based-tools-policies) hereby declare the following regarding the usage of AI tools in the development of the contents of this assignment:

1a) ChatGpt

1b) Purpose of Use: Image generation, Rewriting the paragraphs to improve fluency and proofreading.

1c) The story board was generated by using prompts in ChatGpt.

1d) Design fiction artifact "Postcard from the future" was generated by using prompts in ChatGpt.

We/I declare that no other AI assistance or tools were used in the development of the contents of this assignment than the above mentioned.

Table of Contents

1. Introduction 1

2. Topic Overview: Freebike AI-Powered Customer Support 1

2.1 Problem Definition 1

2.2 Proposed Solution 2

3. UX Methods and Artefacts 3

3.1 Personas and Storyboard (Recap & Critique) 3

3.2 Design Fiction Artefact: Postcard from the Future 4

3.3 Accessible Persona 7

4. Prototyping: 8

4.1 Selected Prototype Screens and Their Derivations 8

5. Evaluation and User Testing 17

5.1. Evaluation Methodology 17

5.2. Results and Findings 19

6. Conclusion 20

7. References 21

8. Appendix: Survey Questionnaires, Test Data & Artifacts 22

Appendix A: SUS Questionnaire Survey and Raw Data 22

Appendix B: Overall User Testing Data 24

Appendix C: Personas and Storyboard 26

1. Introduction

In today's digital world, businesses need to provide quick and effective service in order to keep their customers satisfied. This portfolio details our strategy for improving Freebike's customer service by incorporating an AI chatbot into the Freebike mobile app.

Our first step consisted of a brainstorming session dedicated to identifying the problems encountered by Freebike e-bike users. We covered central issues like the problem description, how an intelligent system could intervene, and what UX artefacts would be able to serve the users' needs adequately. These sessions set the foundation for creating a solution for aiding in a timely manner and helped in making design choices.

The rest of this report presents a comprehensive UX design portfolio for the Freebike AI chatbot project. First, we start with a brief overview of the problem and the proposed solution and then the UX design artifacts used (personas, storyboards, and an advanced method in design fiction), and evidence of designing for marginalized groups via inclusive personas. We then showcase the prototype with interface mockups, illustrating how those insights informed design features. Finally, we detail the evaluation process including user testing methodology, sample evaluation materials, and results and reflect on how well the design meets the UX goals and where improvements might be needed.

2. Topic Overview: Freebike AI-Powered Customer Support

2.1 Problem Definition

Freebike (Mankeli) e-bike users often have to deal with bike technical issues or subscription challenges. Currently, the users have to sit and passively wait for a help ticket to be attended by the customer service. The help ticket system is overly simplistic and does not, most of the time, ensure a quick response. This delay results in anxiety and confusion which reduces overall trust in the service.

The significant user concerns consist of:

- Users want a support system which will help them reach out to during times of emergencies and receive immediate assistance.
- Users want the support process to be straightforward and universal for all user groups, regardless of their language or technical expertise.
- Users want a support experience that complements the current user interface of the application, rather than one that feels different from the existing experience of using the app.
- Users want reassurance, a sense of trust, and confidence that their issue will be resolved when the support service is addressing the problem.

By focusing on these issues from the user's perspective, we will develop solutions that genuinely reflect the needs and emotions of Freebike users.

2.2 Proposed Solution

Our proposed solution is an integrated real-time AI-powered support chatbot that operates within the Freebike mobile application. As a support assistant, this AI chatbot will assist users in real-time with bike breakdowns, subscription issues, billing issues, and reports of lost or damaged bikes. To achieve this, we aim to make the interface as simple and user-friendly as possible to address the concerns of the users.

Users should be able to interact with the AI-powered chatbot directly from the main screen of the app, making it readily visible, rather than having to navigate through different menus to find assistance, which can increase frustration in moments of need. Users should be able to express their issues and receive appropriate answers through the context-aware response feature. Additionally, the system will have an option to provide rating feedback regarding the service.

3. UX Methods and Artefacts

To create this solution, we applied various UX design techniques and created different artifacts throughout the process. The main artifacts include personas, storyboards, and an advanced method - a design fiction scenario - to extend our idea beyond the current context and build a bridge from the future to the present. We also placed special emphasis on inclusive design, making sure to consider marginalized user groups via accessible personas. These methods come from human-centered design and usability engineering practices, and each played a role in shaping our chatbot's design. Below, we detail each method, explaining why we chose it, the stage of design it corresponded to, how it helped us move forward (or its limitations, if any), and the insights - particularly UX requirements in the form of user stories - that we derived from it.

3.1 Personas and Storyboard (Recap & Critique)

In our preliminary research phase, we have created detailed personas and storyboards, which greatly informed our design strategy. Our personas, "John," an international student, and "Taylor," a tourist, allowed us to understand different user needs and pain points of Freebike's customer care services. These artifacts helped us empathize with our target audience, enabling us to make design decisions that effectively address their needs for simplification, speed, and ease of use.

The storyboard illustrated the user journey, depicting how a user encounters a problem, such as a flat tire or technical malfunction, and then seeks chatbot support. The storyboard captured key moments of confusion, anxiety, and relief, emphasizing the need for clear instructions, quick action, and reassurance.

From these artifacts, we were able to extract a few fundamental user expectations:

- Users need a visible, instantly accessible support option.
- Simple step-by-step instructions, which are very important during emergency situations.
- Users want clear, confident messages to reduce stress and anxiety.
- A logical sequence of actions - from reporting an issue to receiving the next steps - is important for maintaining user control and enabling clear troubleshooting.

The personas and storyboard were useful, but as part of our critique, we noted some gaps. Most notably, the primary persona assumptions lacked a more detailed secondary user behavioral analysis, as well as other user narrative aspects, which sometimes omitted subtler components of user emotion and context. The insights gained from these gaps were sufficient to iteratively improve the design, ensuring that later prototype screens directly address the uncovered user challenges. Complete persona details and storyboard visuals are provided in the appendix C for further reference.

3.2 Design Fiction Artefact: Postcard from the Future

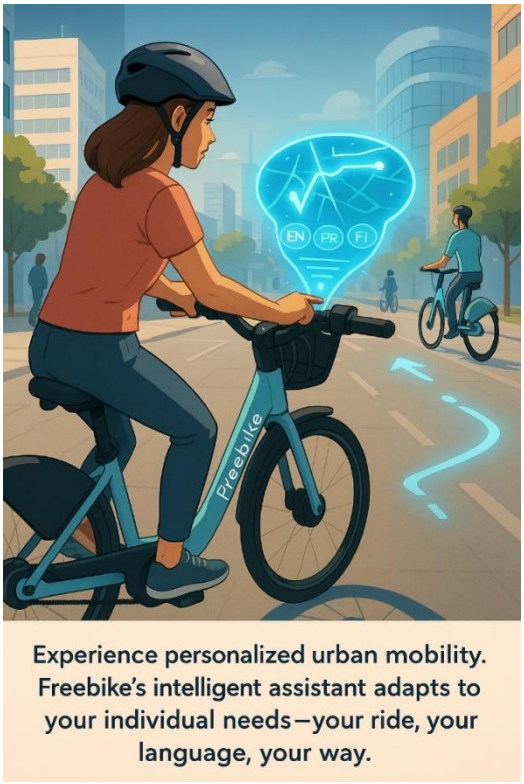


Figure 1. Design Fiction Postcard, Front Side.

Back side of the postcard (text from a future user perspective):

"Hei from Lahti, 2030! I took my grandparents out to enjoy the sunny lakeside yesterday. They've always found using shared e-bikes intimidating, but things have really changed since Freebike upgraded their AI assistant.

Now, the assistant instantly recognizes user profiles and auto-adjusts bike settings: Grandpa got voice-based directions in Finnish (he never liked staring at screens), and Grandma's bike automatically lowered to a comfortable height. The chatbot even remembered from our last ride that Grandma prefers quieter, more scenic paths away from traffic.

It's reassuring to see them confidently enjoying a service they once felt excluded from. But I still wonder - how much does the AI assistant know about us, and how safely is our information handled? I hope they stay transparent and respectful about our data, because this personalized touch makes our days brighter."

We chose Design Fiction ("Postcard from the Future") as one of our advanced UX methods because it effectively allows for the exploration of future user scenarios that current UX methods might overlook. Given that our initial design stage was primarily focused on empathizing and ideating (established through previous personas and storyboard), we selected this method to move us forward to more concrete and detailed prototyping. It encouraged us to creatively visualize potential long-term interactions, pushing our team's understanding beyond present limitations, and fostering an environment where innovative yet grounded ideas could emerge.

Critically evaluating this method, Design Fiction was highly effective in stimulating forward-looking discussions, making it easy to envision how marginalized groups might interact with the technology. However, one limitation is its inherently speculative nature. Without validation through further user research, some envisioned features could remain hypothetical. Despite this potential drawback, it served as an excellent bridge to the prototyping stage by providing tangible concepts for development.

Several insights arose from using this method. It became evident that users highly value personalization, adaptability, and inclusivity in mobility services. Multilingual voice interaction and chat emerged as crucial for non-native speakers, enhancing comfort and accessibility.

Based on these insights, we derived the following UX and usability requirements:

- Elderly or visually impaired users benefit significantly from voice-based navigation and feedback, reducing dependence on visual interfaces.
- Non-native language speakers require multilingual support, accessible through intuitive, effortless interactions.
- All users express strong preferences for clear and transparent data management policies, fostering trust and confidence in the AI assistant.

These specific requirements now directly guide our prototyping and evaluation, ensuring our solutions remain inclusive, intuitive, and accessible to all potential user groups.

3.3 Accessible Persona

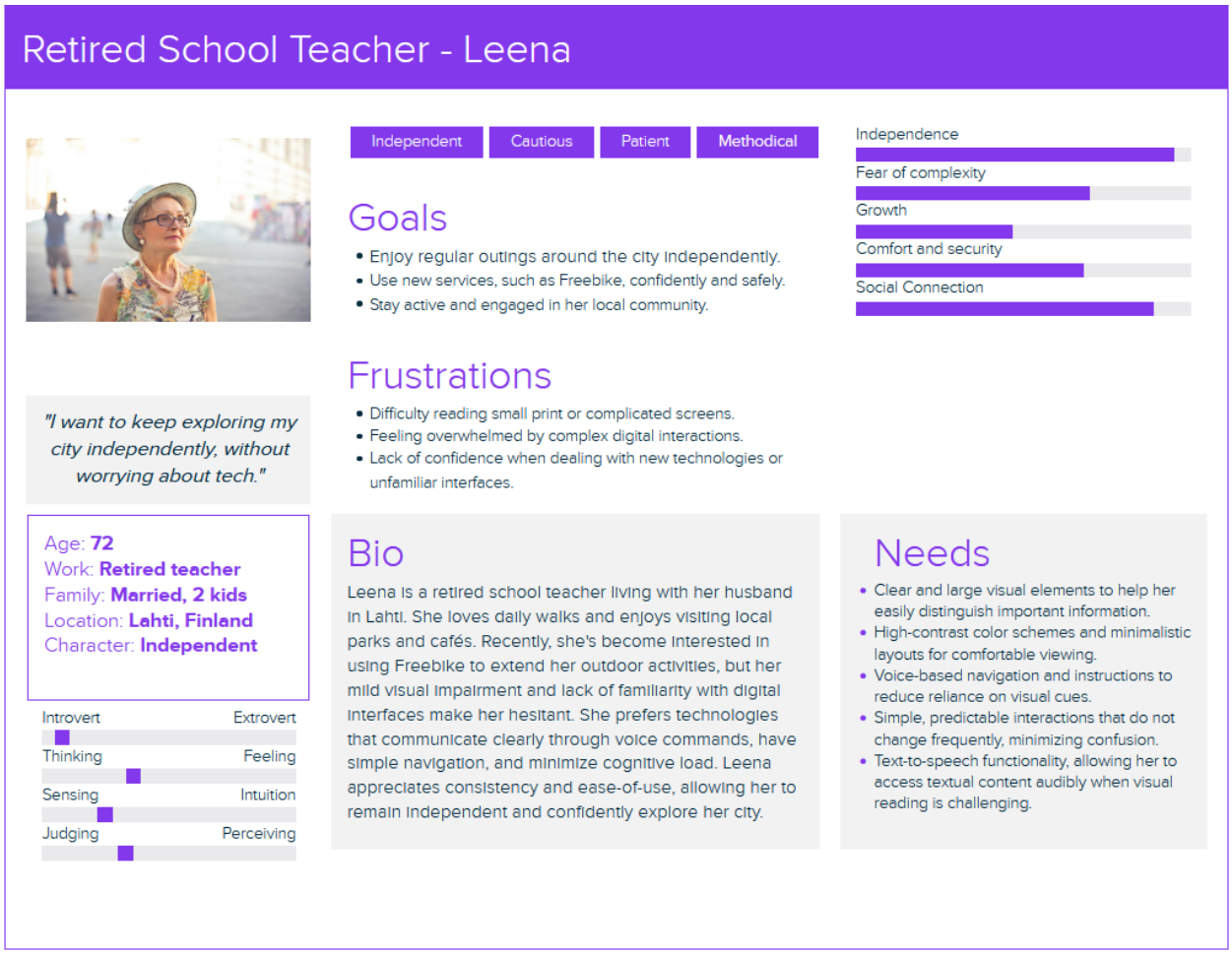


Figure 2. Accessible Persona of Leena

We chose the Accessible Persona method because it provides specific insights into the unique accessibility needs of elderly users, who often experience sensory or cognitive challenges when interacting with digital technology. At the time of selecting this method, we had completed initial ideation phases using traditional personas and storyboarding techniques. The accessible persona method effectively transitioned our design thinking from general user needs to more nuanced considerations for inclusivity, enabling us to move confidently into a more detailed prototyping stage.

Critiquing this method, we found Accessible Personas particularly powerful because they help humanize abstract accessibility challenges by focusing our empathy on the individual rather than on disability

categories. This significantly enhanced our design discussions and decision-making processes. However, a limitation of this method is that without additional user research or real-world validation, assumptions about specific accessibility needs might inadvertently remain generalized.

Using the accessible persona of Leena provided valuable insights, highlighting the necessity for simplified interactions, voice-first interfaces, clear Finnish-language support, and minimalistic visual elements. These insights directly informed several UX and usability requirements:

- **Voice-Guided Interaction:** *"As an elderly user with mild visual impairment, I want clear, easy-to-follow voice-based instructions in my native language so that I can confidently use the service without relying on visual prompts."*
- **Simplified and Consistent Navigation***: "As a user who finds digital interactions challenging, I want simplified menu structures and minimal steps to achieve my goals, enabling me to feel independent and capable."*
- **Enhanced Visual Clarity:** *"As a visually impaired user, I require large, clear icons and text for easy identification, allowing me to interact comfortably with the service."*
- **Consistency and Reliability:** *"As an elderly user, I prefer consistent interaction patterns so that I can quickly learn, remember, and use the interface without confusion or frustration."*

4. Prototyping:

Using the insights and gathered requirements from the personas, storyboard, and design fiction, we continued to create a prototype of the Freebike AI chatbot interface in Figma. The prototype is a mid-fidelity functional mockup of the mobile app's support chat, illustrating the key screens and interactions. Our goal with the prototype was to visibly demonstrate how the earlier UX artifacts translated into design decisions, each major feature in the prototype can be traced back to a user need or requirement identified in our research. In this section, we showcase the prototype (with screenshots) and describe how the design fulfills the UX/usability requirements we set, including those for marginalized users.

4.1 Selected Prototype Screens and Their Derivations

4.1.1 Main App Home with Support Entry Point

This screen displays the Freebike app's home interface with a prominently placed "Support Chatbot" icon.

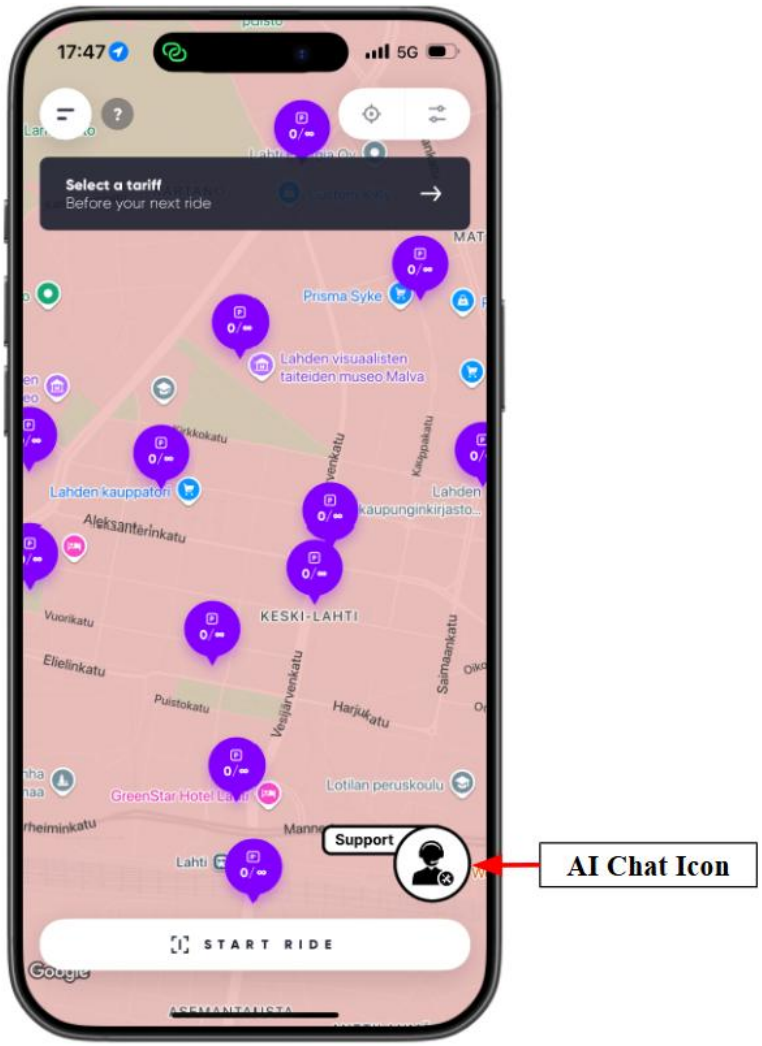


Figure 3: Main App Home with Support Entry Point

Insights:

- Our user personas, especially the international student John, highlighted that immediate, unambiguous access to support is critical. Additionally, storyboarding supported the idea that users should not have to search for help in a complex menu.
- In this screen, the need for immediate assistance and ease of navigation was addressed.

4.1.2 Chatbot Conversation – Reporting an Issue

On this screen, the user initiates a conversation with the chatbot by reporting a problem. The chatbot then asks for additional details to refine the context and also displays transparent data management policies while interacting with the AI.

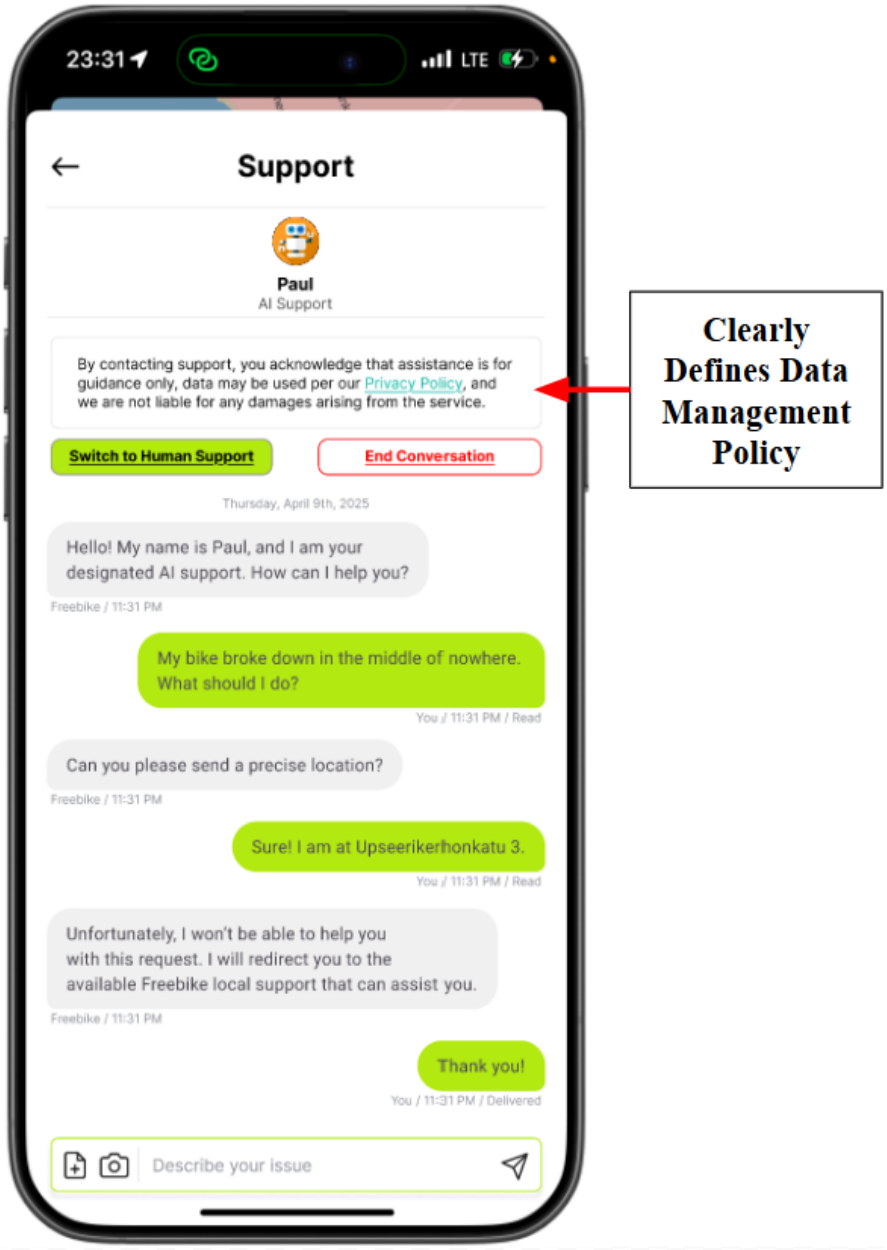


Figure 4: Chatbot Conversation – Reporting an Issue

Insights:

- The storyboard scenario identified the need for clear, step-by-step interaction when issues arise. Additionally, the "Postcard from the Future" emphasized the importance of transparent data management policies while interacting with AI.
- This feature directly addresses the user requirement for clarity in communicating an emergency, as noted in the initial research.

4.1.3 Handoff to Human Support (Kimmo)

On this screen, the chatbot hands the conversation over to a human support agent because of the complexity of the problem. The human assistant, Kimmo, enters the chat and immediately greets the user, verifying their location and informing them that a bike will be sent within minutes.

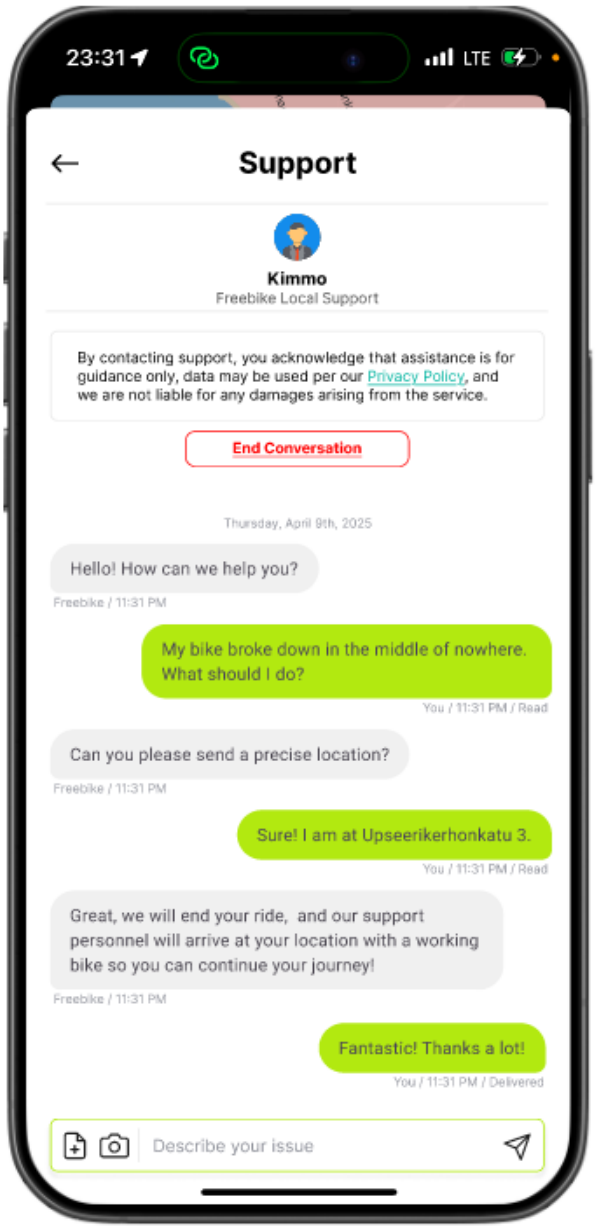


Figure 5: Handoff to Human Support

Insights:

- The “Postcard from the Future” design fiction highlighted that users need reassurance and clear subsequent actions in stressful situations. Our personas endorsed this requirement by emphasizing the need for clear, affirmative responses.
- Reduction of stress and ensuring users feel reassured and in control.

4.1.4 Language Selection

On this screen, the user can choose their desired language for support interaction to begin (Finnish, English, or Swedish). This is presented early in the flow, allowing users to proceed effortlessly in the preferred language.

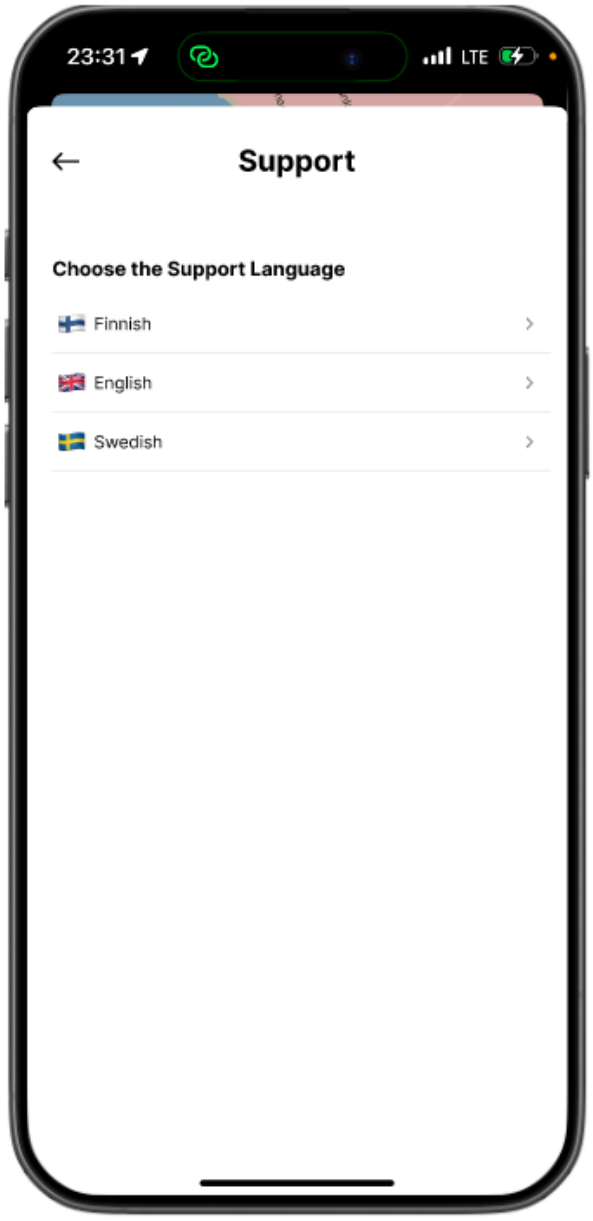


Figure 6: Language Selection

Insights:

- This screen is based on the persona research of John (the international student) and Taylor (the tourist), who clearly expressed their desire to follow instructions in their preferred language.
- It addresses the UX challenge of inclusivity and accessibility for foreign languages by providing a non-native speaker-friendly interface, thus broadening the user base for Freebike.

4.1.5 Accessible Interaction Mode

This screen is optimized for accessibility with high-contrast visuals, larger text, and a visible voice-input button, allowing users to report issues via speech.

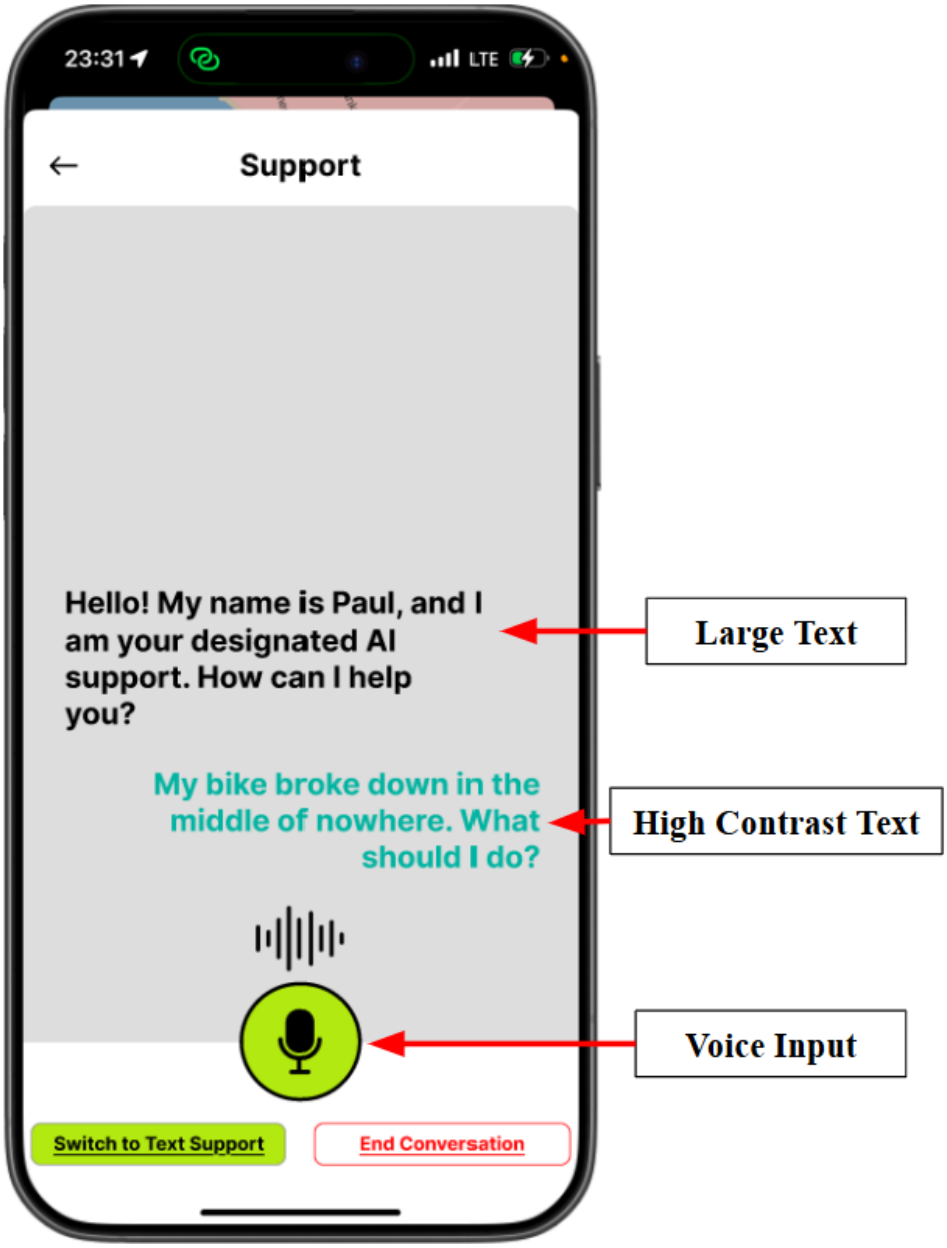


Figure 7: Accessible Interaction Mode

Insights:

- Our Accessible Persona, “Leena,” stressed the necessity of voice interaction and enhanced visual elements for users with visual impairments or motor difficulties.
- Inclusive design ensures that the interface is accessible and user-friendly for everyone.

4.1.6 Feedback and Rating Screen

After the support interaction, this screen invites users to rate their experience using a simple star system.

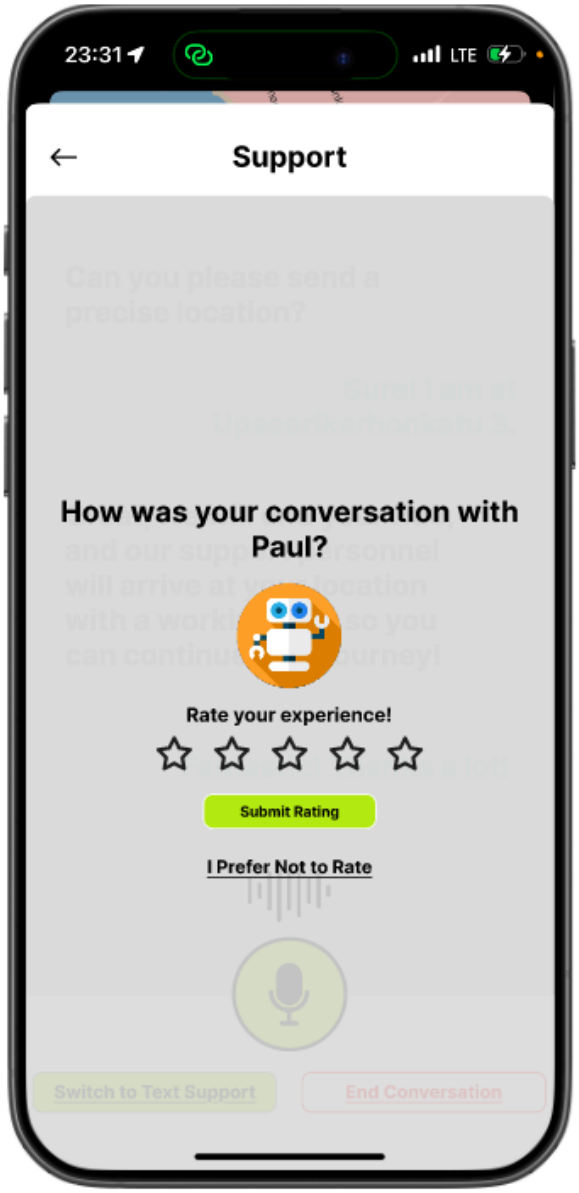


Figure 8: Feedback and Rating Screen

Insights:

- Feedback was a core requirement identified during our initial research and was reinforced through our storyboarding and persona sessions as essential for continuous improvement.
- This involves transparent feedback collection and ongoing service enhancement.

5. Evaluation and User Testing

After building the prototype, we moved to the evaluation phase to test the usability and effectiveness of the Freebike AI chatbot with real users. Our evaluation strategy combined a usability test with representative users, a post-test survey (SUS questionnaire) to quantitatively evaluate usability, and a short follow-up interview for qualitative feedback. The goals of the evaluation were to validate that the design meets the UX goals (immediate assistance, ease of use, stress reduction, etc.), to uncover any usability problems, and to see how the chatbot performs for different user groups (including those from our personas). We also wanted to reference our earlier requirements to see if they were fulfilled in practice. For example, did users find the support feature easily? Did they feel reassured and in control?

5.1. Evaluation Methodology

We recruited six participants reflecting our persona profiles and beyond: two international students (similar to John), two local Finnish users in their 20s (to understand the local perspective), one tourist (a friend visiting, serving as a proxy for Taylor), and one older user with mild visual impairment (serving as a proxy for Leena). This diverse set allowed us to test the chatbot with marginalized considerations in mind. Jakob Nielsen's research suggests that testing with five users can uncover the majority of usability issues (Nielsen, 2000), and we aimed slightly above that number to include the older user for accessibility insights.

Each participant was given a smartphone with the Freebike chatbot prototype installed in the Figma app. We conducted the tests in one-on-one sessions. The test with the older user was done in person with accommodations, using their own phone with larger display settings.

We created a task scenario for users to perform, mirroring the storyboard scenario but updated to reflect the current prototype's features as follows:

- From the Home screen, users look for and press the "Support" button on the lower right side of the screen. They then see the "Choose the Support Language" screen and select the appropriate language.

- Participants then specify whether they wish to chat with the AI support or use the voice support option. They describe their problem (e.g., “My bike has broken down”) either by text or via voice input using the voice-assisted feature. They follow successive prompts to provide any additional details.
- They review the details provided from the support messages and, if necessary, ask for more information. In case of insufficient details, they can switch to person-operated support. Once all the messages are reviewed and satisfactory, both text and voice chats can be ended.
- At the end, participants navigate to the feedback and rating screen to submit their ratings and finish the conversation.

We used a think-aloud protocol, participants were instructed to voice their thoughts, expectations, or confusion as they interacted with the chatbot. We observed and took notes. We timed the tasks to see how quickly they could get a resolution.

After the task, participants filled out the System Usability Scale (SUS) questionnaire (Brooke, 1996). The SUS is a standard 10-item Likert scale survey for subjective usability assessment. It includes statements such as “I thought the system was easy to use” and “I felt very confident using the system,” rated from Strongly Disagree (1) to Strongly Agree (5). We chose the SUS because it’s quick and provides a single score out of 100 that is easy to interpret, with 68 considered as average usability score (MeasuringU, n.d.). Additionally, we added five custom questions to the survey to target our specific UX goals, such as:

- Q1. "*Did the chatbot provide you with help quickly enough?*" (Not at all – Very much so, 5-point scale)
- Q2. "*How easy was it to understand the chatbot’s instructions?*" (Very difficult – Very easy, 5-point scale)
- Q3. "*Did you trust the answers the chatbot gave you?*" (Not at all – Completely, 5-point scale)
- Q4. "*How satisfied are you with the support experience overall?*" (1-5 stars)
- Q5. "*Would you prefer this chatbot over waiting for human support via email?*" (Yes/No, and please explain why)

Finally, we held a brief interview (~10 minutes) with each participant, asking open-ended questions such as:

- "*What did you like or dislike about the support chatbot?*"
- "*Was there anything confusing or frustrating?*"
- "*Do you have any suggestions to improve it?*"

We also asked the older user, "*How was the experience using voice, and did the interface feel accessible to you?*" Additionally, we tailored questions based on each user's experience. For example, we asked one student who initially struggled to find the support button what they had expected.

Appendices A and B include the full list of survey questions, and the test script used for the sessions.

5.2. Results and Findings

The evaluation yielded several valuable insights:

- The chatbot received a high average SUS score of 85 out of 100, which is well above average, indicating that users found it easy and pleasant to use. Comments included, "It was surprisingly simple," and "Way better than expected."
- Users completed tasks quickly, often in under two minutes. All users confirmed that they received help promptly, meeting our goal of providing immediate assistance.
- Most users easily located the chatbot on the main screen. However, one participant initially missed the chatbot icon, expecting to find support in a menu instead. This suggests that we could slightly improve visibility or add a brief introductory hint.
- The instructions provided by the chatbot were generally easy to follow, receiving a score of 4.8 out of 5. Users particularly appreciated the step-by-step guidance.
- If the AI is unable to assist, it directly connects users to human support based on the severity of the problem. All users felt reassured by this, which helped reduce stress and significantly contributed to user trust.
- International participants successfully used the multilingual features and found the language options intuitive. Our older participant effectively used voice commands, although there are opportunities for minor improvements in accessibility features.

Based on the evaluation, we identified the following improvements for the next iteration:

- Add clear visual cues or introductory tips to ensure the chatbot icon is immediately noticeable to all users.
- Clarify certain instructions.
- Further refine voice-command usability and visual clarity to enhance accessibility.

6. Conclusion

This portfolio reflects our complete UX design workflow for the Freebike AI chatbot, developed as a support tool for an e-bike sharing service. The working prototype is user-friendly, featuring a responsive, context-aware, and highly interactive chat interface. Testing with users confirmed that participants received rapid support and effective directions, allowing them to achieve their goals swiftly. The warm tone and fast responses were also praised in the feedback. Overall, the response demonstrates that our focus on users' needs during the design and research phases streamlined customer support with increased efficiency and simplicity. In summary, our approach successfully addressed critical user requirements. With advancing technology and increasing demands, the chatbot will continue to be improved to ensure it remains a reliable support resource.

7. References

Brooke, J., 1996. SUS: A “quick and dirty” usability scale. In: P.W. Jordan, B. Thomas, B.A. Weerdmeester and A.L. McClelland, eds. Usability Evaluation in Industry. London: Taylor & Francis, pp.189–194.

Nielsen, J. (2000) *Why You Only Need to Test with 5 Users*. Nielsen Norman Group. Available at: https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/ (Accessed: 10 April 2025).

Wikipedia contributors (n.d.) *System Usability Scale*. Wikipedia, The Free Encyclopedia. Available at: https://en.wikipedia.org/wiki/System_Usability_Scale (Accessed: 07 April 2025).

MeasuringU (n.d.) *What is SUS?*. MeasuringU. Available at: https://measuringu.com/sus/ (Accessed: 07 April 2025).

8. Appendix: Survey Questionnaires, Test Data & Artifacts

Appendix A: SUS Questionnaire Survey and Raw Data

A.1 SUS Questionnaire

Item	Statement	1 Strongly Disagree	2	3	4	5 Strongly Agree
1	I think that I would like to use this system frequently.					
2	I found the system unnecessarily complex.					
3	I thought the system was easy to use.					
4	I think that I would need the support of a technical person to be able to use this system.					
5	I found the various functions in this system were well integrated.					
6	I thought there was too much inconsistency in this system.					
7	I would imagine that most people would learn to use this system very quickly.					
8	I found the system very cumbersome to use.					
9	I felt very confident using the system.					
10	I needed to learn a lot of things before I could get going with this system.					

A.2 Feedback data

Participant	Q1 (Pos)	Q2 (Neg)	Q3 (Pos)	Q4 (Neg)	Q5 (Pos)	Q6 (Neg)	Q7 (Pos)	Q8 (Neg)	Q9 (Pos)	Q10 (Neg)	Final SUS
P1	5	1	5	2	4	2	5	2	5	2	88
P2	4	1	5	2	4	2	5	2	5	2	85
P3	4	2	5	2	4	2	5	2	5	2	82
P4	5	2	4	2	5	2	4	1	5	2	86
P5	5	2	5	2	4	4	5	2	5	3	78
P6	5	1	5	1	5	1	5	3	5	3	90

SUS Calculation Method:

1. The **System Usability Scale** (SUS) has **10 items** (Q1–Q10).
2. Odd numbered items (Q1, Q3, Q5, Q7, Q9) are **positively worded**.
    - **Adjusted value** = (Rating – 1).
3. Even numbered items (Q2, Q4, Q6, Q8, Q10) are **negatively worded**.
    - **Adjusted value** = (5 – Rating).
4. Sum the 10 adjusted values (range: 0–40).
5. Multiply by 2.5 to get a final SUS score (range: 0–100).

Appendix B: Overall User Testing Data

B.1 Quantitative Data

Participant	User Type	SUS Score	Time to Complete Task (sec)	Help Speed (1–5)	Clarity (1–5)	Trust (1–5)	Overall Satisfaction (1–5)
P1 (John-type)	Intl. Student	88	98	5	5	4	5
P2 (Taylor-type)	Tourist	85	120	4	5	5	4
P3 (Local)	Commuter	82	110	4	4	4	4
P4 (Local)	Commuter	86	105	5	4	4	4
P5 (Leena-type)	Visually Imp.	78	140	4	4	3	4
P6 (Local)	Commuter	90	95	5	5	4	5

Averages:

- SUS: 84.8
- Time to Complete: 111 seconds
- Help Speed: 4.5 / 5
- Clarity: 4.5 / 5
- Trust: 4.0 / 5
- Overall Satisfaction: 4.3 / 5

B.2 Feedback (Representative Quotes)

- **P1 (Intl. Student)**: “It responded instantly… I really liked how it told me not to worry about extra fees.”
- **P2 (Tourist)**: “The instructions were super clear, but I was a bit confused finding the help button at first.”
- **P5 (Visually Impaired)**: “I could read the text mostly fine but would love a voice response option too.”
- **P6 (Local)**: “I trust it because it explains why it picked that station for me.”

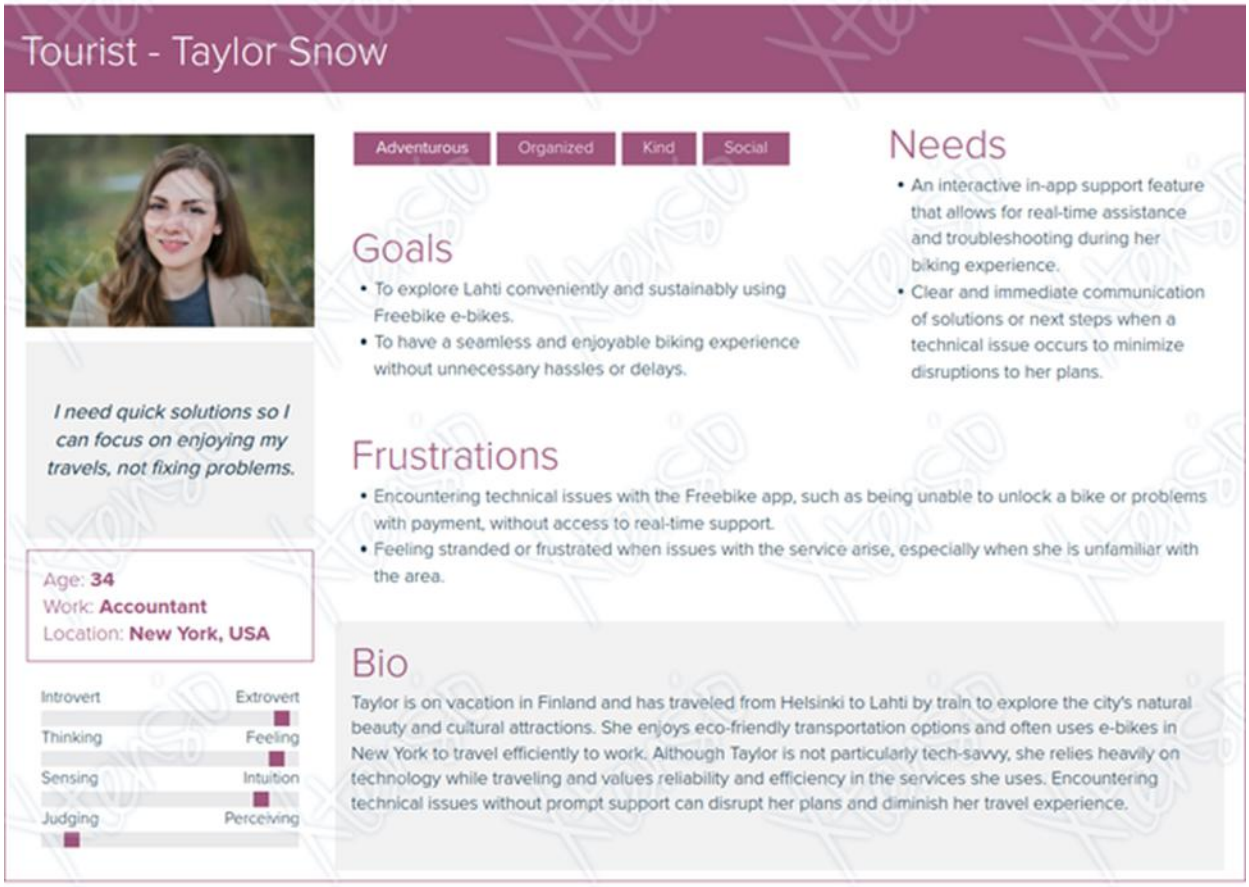
Appendix C: Personas and Storyboard

C.1 Personas

International Student – Jhon Smith



Tourist – Taylor Snow



C.2 Storyboard

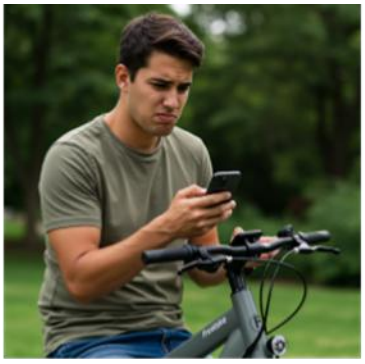
Storyboard of John Smith using the new chatbot

1.



John Smith was riding a Mankeli bike to the university and it suddenly broke. The bike malfunctions! John is stuck and frustrated.

2.



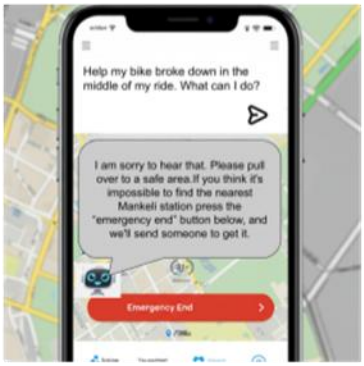
Now needing to deal with this, he is wondering how things will work out. Will this be another slow, unhelpful support experience?

3.



John opens the Freebike app and the new AI chatbot pops up to offer immediate support—no waiting required.

4.



The AI provides clear, actionable steps to help the user resolve the issue quickly, ensuring responsiveness and swift support. Interactive and responsive support for is a nice surprise for John

5.



John's mind is at peace and and he is relieved. Thanks to the chatbot he knows he can leave the bike here and the management will deal with it. All he has to do is to go get a new bike!

6.



John goes to the nearest station to unlock a new bike in good spirits. He doesn't need to worry about future problems with customer support issues.